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Thermodynamics and Statistical Physics

Spring 2026

Homework 8

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Show all essential steps in your derivations.

Composition variable in a binary mixture ⋆⋆

Consider a binary mixture consisting of two species, with particle numbers N_1 and N_2, and total particle number

$$N = N_1 + N_2.$$

The differential form of the Helmholtz free energy is [c.f. Table **??**]

$$dF = -S\,dT - P\,dV + \mu_1\,dN_1 + \mu_2\,dN_2.$$

Define the mole fraction of species 1 by

$$x = \frac{N_1}{N}, \qquad 1 - x = \frac{N_2}{N},$$

and introduce the intensive variables

$$v = \frac{V}{N}, \qquad s = \frac{S}{N}, \qquad f = \frac{F}{N}.$$

By extensivity, the Helmholtz free energy may be written as

$$F(T, V, N_1, N_2) = N f(T, v, x), \qquad v = \frac{V}{N}, \qquad x = \frac{N_1}{N}.$$

Thus f may be regarded as a function

$$f = f(T, v, x).$$

1. Show that

$$df = -s\,dT - P\,dv + (\mu_1 - \mu_2)\,dx.$$

   Hence

$$\left(\frac{\partial f}{\partial x}\right)_{T,v} = \mu_1 - \mu_2.$$

2. Define the Gibbs free energy

$$G = F + PV.$$

   Using Euler's relation for a multicomponent system,

$$G = \mu_1 N_1 + \mu_2 N_2,$$

   show that

$$f = -Pv + x\mu_1 + (1 - x)\mu_2.$$

3. Combine the preceding results to show that

$$\begin{aligned}
\mu_2 &= f + Pv - x\left(\frac{\partial f}{\partial x}\right)_{T,v}, \\
\mu_1 &= f + Pv + (1 - x)\left(\frac{\partial f}{\partial x}\right)_{T,v}.
\end{aligned}$$

Solution

1. We start with the extensivity of the Helmholtz free energy:

$$F(T, V, N_1, N_2) = N f(T, v, x)$$

Taking the total differential of $F = Nf$, we get:

$$dF = N\,df + f\,dN \tag{1}$$

We are also given the standard thermodynamic identity for dF:

$$dF = -S\,dT - P\,dV + \mu_1\,dN_1 + \mu_2\,dN_2 \tag{2}$$

We need to express dV, dN_1, and dN_2 in terms of dv, dx, and dN. Since $v = V/N$, $x = N_1/N$, and $1 - x = N_2/N$, we have:

$$\begin{aligned} V = Nv &\implies dV = N\,dv + v\,dN \\ N_1 = Nx &\implies dN_1 = N\,dx + x\,dN \\ N_2 = N(1-x) &\implies dN_2 = -N\,dx + (1-x)\,dN \end{aligned}$$

Substitute these differentials into Eq. (2):

$$\begin{aligned} dF &= -S\,dT - P(N\,dv + v\,dN) + \mu_1(N\,dx + x\,dN) + \mu_2(-N\,dx + (1-x)\,dN) \\ &= -S\,dT - PN\,dv + N(\mu_1 - \mu_2)\,dx + [-Pv + x\mu_1 + (1-x)\mu_2]\,dN \end{aligned}$$

Equating this to Eq. (1) and substituting $S = Ns$:

$$N\,df + f\,dN = -Ns\,dT - PN\,dv + N(\mu_1 - \mu_2)\,dx + [-Pv + x\mu_1 + (1-x)\mu_2]\,dN$$

Since $f(T, v, x)$ is an intensive variable independent of N, its differential df can only contain dT, dv, and dx. By matching the terms proportional to N, we obtain $N\,df$:

$$N\,df = -Ns\,dT - PN\,dv + N(\mu_1 - \mu_2)\,dx$$

Dividing both sides by N gives the desired differential:

$$df = -s\,dT - P\,dv + (\mu_1 - \mu_2)\,dx$$

From this exact differential, we can immediately read off the partial derivative with respect to x:

$$\left(\frac{\partial f}{\partial x}\right)_{T,v} = \mu_1 - \mu_2$$

2. From Euler's relation for a multicomponent system, the Gibbs free energy is:

$$G = \mu_1 N_1 + \mu_2 N_2$$

By definition, $G = F + PV$. Equating the two expressions for G and dividing by the total particle number N:

$$\frac{F}{N} + P\frac{V}{N} = \mu_1\frac{N_1}{N} + \mu_2\frac{N_2}{N}$$

Substituting the intensive variables $f = F/N$, $v = V/N$, $x = N_1/N$, and $1 - x = N_2/N$, we get:

$$f + Pv = x\mu_1 + (1-x)\mu_2$$

Rearranging for f yields the required result:

$$f = -Pv + x\mu_1 + (1-x)\mu_2$$

(Note: This is exactly the coefficient of dN derived in Part 1, confirming $fdN = [-Pv + x\mu_1 + (1-x)\mu_2]dN$.)

3. We now have a system of two equations for the two unknowns μ_1 and μ_2:

$$\mu_1 - \mu_2 = \left(\frac{\partial f}{\partial x}\right)_{T,v} \tag{3}$$

$$x\mu_1 + (1-x)\mu_2 = f + Pv \tag{4}$$

To solve for μ_2, we substitute $\mu_1 = \mu_2 + (\partial f/\partial x)_{T,v}$ from (3) into (4):

$$\begin{aligned} x\left[\mu_2 + \left(\frac{\partial f}{\partial x}\right)_{T,v}\right] + (1-x)\mu_2 &= f + Pv \\ \mu_2 + x\left(\frac{\partial f}{\partial x}\right)_{T,v} &= f + Pv \\ \mu_2 &= f + Pv - x\left(\frac{\partial f}{\partial x}\right)_{T,v} \end{aligned}$$

Similarly, to solve for μ_1, we substitute $\mu_2 = \mu_1 - (\partial f/\partial x)_{T,v}$ from (3) into (4):

$$\begin{aligned} x\mu_1 + (1-x)\left[\mu_1 - \left(\frac{\partial f}{\partial x}\right)_{T,v}\right] &= f + Pv \\ \mu_1 - (1-x)\left(\frac{\partial f}{\partial x}\right)_{T,v} &= f + Pv \\ \mu_1 &= f + Pv + (1-x)\left(\frac{\partial f}{\partial x}\right)_{T,v} \end{aligned}$$

Thus, the relations are proven.

Mixing of two gases with unequal pressures ⋆⋆⋆

Suppose two ideal gases of different species are initially separated by a partition. They have the same temperature T, but have particle numbers N_1, N_2 and volumes V_1, V_2, with

$$\frac{N_1}{V_1} \neq \frac{N_2}{V_2}.$$

Thus the two gases are initially in thermal equilibrium, but not in mechanical equilibrium. After removing the partition, the gases mix and occupy the total volume

$$V = V_1 + V_2.$$

Since the gases are ideal and have the same initial temperature, the final temperature is still T.

1. Starting from the free-energy change

$$\Delta F = T \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V_\alpha}{V}\right),$$

show that the total entropy change is

$$\Delta S = \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V}{V_\alpha}\right). \tag{5}$$

2. Define

$$N = N_1 + N_2, \qquad x_\alpha = \frac{N_\alpha}{N}, \qquad n_{\alpha,\mathrm{init}} = \frac{N_\alpha}{V_\alpha}, \qquad n_{\mathrm{final}} = \frac{N}{V}.$$

Show that Eq. (5) can be rewritten as

$$\Delta S = \Delta S_{\mathrm{mix}} + \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\mathrm{init}}}{n_{\mathrm{final}}}\right), \tag{6}$$

where

$$\Delta S_{\mathrm{mix}} = -\sum_{\alpha=1}^{2} N_\alpha \log x_\alpha. \tag{7}$$

3. Show that the second term on the right-hand side of Eq. (6) is nonnegative as a whole:

$$\sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\mathrm{init}}}{n_{\mathrm{final}}}\right) \geq 0. \tag{8}$$

Also show that it vanishes when the two initial densities are equal.

Hint: First show that

$$\frac{1}{n_{\mathrm{final}}} = \sum_{\alpha=1}^{2} x_\alpha \frac{1}{n_{\alpha,\mathrm{init}}},$$

or equivalently,

$$1 = \sum_{\alpha=1}^{2} x_\alpha \frac{n_{\mathrm{final}}}{n_{\alpha,\mathrm{init}}}.$$

Then use the inequality

$$-\log y \geq 1 - y.$$

4. Now consider a gas consisting of a single species of molecules. Initially, the gas is divided into two parts at the same temperature T, containing N_1 and N_2 particles in volumes V_1 and V_2, respectively. The two initial densities are not necessarily equal. The partition is then removed, and the gas relaxes irreversibly to a uniform final state occupying the total volume

$$V = V_1 + V_2,$$

with final density

$$n_{\text{final}} = \frac{N_1 + N_2}{V}.$$

Show that the entropy increase in this single-species process is precisely the second term on the right-hand side of Eq. (6). Here $\alpha = 1, 2$ labels the two initial compartments, not two different species.

5. Explain the physical meaning of the decomposition Eq. (6). In particular, explain why ΔS_{mix} is the genuine entropy of mixing due to the interpenetration of two different species, while the second term is the entropy increase due to the relaxation of an initially nonuniform density profile.

Solution

1. Since the gases are ideal and the initial and final temperatures are the same (T), the total internal energy change of the system is zero ($\Delta U = 0$). Using the thermodynamic relation $F = U - TS$, the change in free energy at constant temperature is:

$$\Delta F = \Delta U - T\Delta S = -T\Delta S$$

We are given:

$$\Delta F = T \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V_\alpha}{V}\right)$$

Equating the two expressions for ΔF:

$$-T\Delta S = T \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V_\alpha}{V}\right)$$

Dividing by $-T$ gives the total entropy change:

$$\Delta S = -\sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V_\alpha}{V}\right) = \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{V}{V_\alpha}\right)$$

2. We want to rewrite V/V_α in terms of densities and fractions. By definition:

$$n_{\alpha,\text{init}} = \frac{N_\alpha}{V_\alpha} \implies V_\alpha = \frac{N_\alpha}{n_{\alpha,\text{init}}}$$

$$n_{\text{final}} = \frac{N}{V} \implies V = \frac{N}{n_{\text{final}}}$$

Substitute these into the volume ratio:

$$\frac{V}{V_\alpha} = \frac{N/n_{\text{final}}}{N_\alpha/n_{\alpha,\text{init}}} = \left(\frac{N}{N_\alpha}\right)\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right) = \frac{1}{x_\alpha}\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}$$

Substitute this back into the entropy change equation:

$$\begin{aligned}\Delta S &= \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{1}{x_\alpha}\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right) \\ &= \sum_{\alpha=1}^{2} N_\alpha \left[-\log x_\alpha + \log\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right)\right] \\ &= -\sum_{\alpha=1}^{2} N_\alpha \log x_\alpha + \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right)\end{aligned}$$

Defining $\Delta S_{\text{mix}} = -\sum_{\alpha=1}^{2} N_\alpha \log x_\alpha$, we arrive at:

$$\Delta S = \Delta S_{\text{mix}} + \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right)$$

3. First, we prove the hint. The total volume is $V = V_1 + V_2$. Dividing by N:

$$\frac{V}{N} = \frac{V_1}{N} + \frac{V_2}{N}$$

Since $V/N = 1/n_{\text{final}}$ and $V_\alpha = N_\alpha/n_{\alpha,\text{init}}$, we get:

$$\frac{1}{n_{\text{final}}} = \frac{N_1/n_{1,\text{init}}}{N} + \frac{N_2/n_{2,\text{init}}}{N} = x_1\frac{1}{n_{1,\text{init}}} + x_2\frac{1}{n_{2,\text{init}}} = \sum_{\alpha=1}^{2} x_\alpha \frac{1}{n_{\alpha,\text{init}}}$$

Multiplying both sides by n_{final} gives:

$$1 = \sum_{\alpha=1}^{2} x_\alpha \frac{n_{\text{final}}}{n_{\alpha,\text{init}}}$$

Now, analyze the second term of the entropy change. Let $y_\alpha = n_{\text{final}}/n_{\alpha,\text{init}}$. We have $\sum x_\alpha y_\alpha = 1$. The term to evaluate is:

$$X = \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right) = -N\sum_{\alpha=1}^{2} x_\alpha \log y_\alpha$$

Using the fundamental inequality $-\log y \geq 1 - y$ for $y > 0$:

$$X \geq N\sum_{\alpha=1}^{2} x_\alpha(1 - y_\alpha) = N\left(\sum_{\alpha=1}^{2} x_\alpha - \sum_{\alpha=1}^{2} x_\alpha y_\alpha\right)$$

Since $\sum x_\alpha = 1$ and we proved $\sum x_\alpha y_\alpha = 1$:

$$X \geq N(1-1) = 0$$

This proves $\sum N_\alpha \log(n_{\alpha,\text{init}}/n_{\text{final}}) \geq 0$. The equality $-\log y = 1 - y$ holds if and only if $y = 1$, which means $y_1 = y_2 = 1$, corresponding to equal initial densities $n_{1,\text{init}} = n_{2,\text{init}} = n_{\text{final}}$.

4. For a single-species ideal gas, the entropy of N particles in volume V at temperature T is $S(N, V) = N \log(V/N) + Nf(T) = -N \log n + Nf(T)$. The initial entropy of the two separated compartments is:

$$S_{\text{init}} = S(N_1, V_1) + S(N_2, V_2) = -N_1 \log n_{1,\text{init}} + N_1 f(T) - N_2 \log n_{2,\text{init}} + N_2 f(T)$$

The final entropy after removing the partition is:

$$S_{\text{final}} = S(N_1 + N_2, V) = -(N_1 + N_2) \log n_{\text{final}} + (N_1 + N_2) f(T)$$

The entropy increase is:

$$\begin{aligned}
\Delta S_{\text{single}} &= S_{\text{final}} - S_{\text{init}} \\
&= -N_1 \log n_{\text{final}} - N_2 \log n_{\text{final}} + N_1 \log n_{1,\text{init}} + N_2 \log n_{2,\text{init}} \\
&= N_1 \log\left(\frac{n_{1,\text{init}}}{n_{\text{final}}}\right) + N_2 \log\left(\frac{n_{2,\text{init}}}{n_{\text{final}}}\right) = \sum_{\alpha=1}^{2} N_\alpha \log\left(\frac{n_{\alpha,\text{init}}}{n_{\text{final}}}\right)
\end{aligned}$$

This is exactly the second term of Eq. (6).

5. **Physical meaning:** The decomposition in Eq. (6) splits the total entropy change into two distinct physical origins. 1. $\Delta S_{\text{mix}} = -\sum N_\alpha \log x_\alpha$ is the **pure entropy of mixing**. It arises purely from the indistinguishability of identical particles vs. distinguishability of different species. It is the information lost when two different species interpenetrate, assuming they started at the same pressure/density. 2. $\sum N_\alpha \log(n_{\alpha,\text{init}}/n_{\text{final}})$ is the **entropy of density relaxation (or mechanical equilibration)**. As shown in part 4, this term exists even for a single species. It arises from the irreversible expansion or compression of the gases from their initial non-equilibrium partial densities to the final uniform equilibrium density. Thus, the total entropy change is the sum of chemical/species mixing and mechanical density homogenization.

Entropy of mixing in an ideal solution ⋆⋆

Consider a classical system at fixed temperature T and volume V, consisting of N_A particles of species A and N_B particles of species B, with

$$N = N_A + N_B.$$

Set $k_B = 1$. Assume that the two species are **ideal-solution components**: they have the same mass and the same interaction energies,

$$u_{AA}(r) = u_{BB}(r) = u_{AB}(r).$$

Therefore the total potential energy depends only on the particle positions, and not on whether a given particle is labeled A or B.
Let the thermal wavelength of either species be denoted by λ_T. Let

$$U(q_1, \ldots, q_N)$$

be the potential energy of the N-particle configuration.

1. Show that the canonical partition function of the binary mixture is

$$Z_{\text{mix}}(N_A, N_B, V, T) = \frac{1}{N_A!\, N_B!\, \lambda_T^{3N}} \int_{V^N} dq_1 \cdots dq_N \, e^{-\beta U(q_1,\ldots,q_N)}.$$

   Explain why the Gibbs factor is $1/(N_A!N_B!)$, rather than $1/N!$.

2. Now consider a reference system consisting of $N = N_A + N_B$ particles of a single species, with the same mass and the same interaction potential. Show that its canonical partition function is

$$Z_{\text{pure}}(N, V, T) = \frac{1}{N!\, \lambda_T^{3N}} \int_{V^N} dq_1 \cdots dq_N \, e^{-\beta U(q_1,\ldots,q_N)}.$$

3. Define the free energy of mixing by

$$\Delta F \equiv F_{\text{mix}} - F_{\text{pure}} = -T \log Z_{\text{mix}} + T \log Z_{\text{pure}}.$$

   Show that

$$\Delta F \quad = \quad -T \log \frac{N!}{N_A!\, N_B!}. \tag{9}$$

   Using Stirling's formula for macroscopic systems, show that

$$\Delta F \quad = \quad T\Big[N_A \log x_A + N_B \log x_B\Big] < 0, \tag{10}$$

   where

$$x_A = \frac{N_A}{N}, \qquad x_B = \frac{N_B}{N}.$$

This result is also the free-energy change for mixing two initially separated liquids of species A and B, provided that the two species have the same mass and identical interactions. Under this condition, mixing does not change the interaction part of the partition function. The entire free-energy change comes from the combinatorial factor counting the possible assignments of A and B labels among the N particle positions.

4. Show that the entropy change associated with this mixing process is

$$\Delta S_{\text{mix}} = -\Big[N_A \log x_A + N_B \log x_B\Big].$$

Equivalently,

$$\Delta F = -T\Delta S_{\text{mix}}.$$

5. Explain why the results apply to dense liquid solutions. Explain why the free-energy change of mixing in an ideal solution is proportional to T, and why the entropy of mixing is independent of T.

Solution

1. In a classical system, the canonical partition function requires integrating the Boltzmann factor $e^{-\beta U}$ over the phase space. Since the N_A particles of species A are fundamentally indistinguishable from one another, we must divide the phase space volume by $N_A!$ to avoid overcounting microstates. Similarly, the N_B particles of species B are indistinguishable among themselves, requiring a division by $N_B!$. However, particles of species A are distinguishable from particles of species B, so we **do not** divide by the total $(N_A + N_B)!$. Both species have the same mass, so they share the same thermal wavelength λ_T. The momentum integrals yield λ_T^{-3N}. Thus, the partition function is:

$$Z_{\text{mix}}(N_A, N_B, V, T) = \frac{1}{N_A!\, N_B!\, \lambda_T^{3N}} \int_{V^N} dq_1 \cdots dq_N \, e^{-\beta U(q_1,\ldots,q_N)}$$

2. For the pure reference system, all $N = N_A + N_B$ particles belong to the exact same species. Therefore, all N particles are mutually indistinguishable. To correct the classical phase space volume for this indistinguishability, the Gibbs factor must be $1/N!$. The masses and interaction potentials are identical to the mixture case. This gives:

$$Z_{\text{pure}}(N, V, T) = \frac{1}{N!\, \lambda_T^{3N}} \int_{V^N} dq_1 \cdots dq_N \, e^{-\beta U(q_1,\ldots,q_N)}$$

3. The free energy of mixing is given by:

$$\Delta F = -T \log Z_{\text{mix}} + T \log Z_{\text{pure}} = -T \log\left(\frac{Z_{\text{mix}}}{Z_{\text{pure}}}\right)$$

Since the interaction potentials $U(q_1, \ldots, q_N)$ are identical (ideal-solution assumption), the configurational integrals in Z_{mix} and Z_{pure} are exactly the same and cancel out upon division. The thermal wavelengths λ_T^{3N} also cancel. We are left purely with the combinatorics:

$$\frac{Z_{\text{mix}}}{Z_{\text{pure}}} = \frac{1/(N_A!\,N_B!)}{1/N!} = \frac{N!}{N_A!\,N_B!}$$

Thus, the exact free energy of mixing is:

$$\Delta F = -T \log \frac{N!}{N_A!\,N_B!}$$

Using Stirling's approximation $\log N! \approx N \log N - N$ for macroscopic systems:

$$\begin{aligned}\log \frac{N!}{N_A!\,N_B!} &= \log N! - \log N_A! - \log N_B! \\ &\approx (N \log N - N) - (N_A \log N_A - N_A) - (N_B \log N_B - N_B)\end{aligned}$$

Since $N = N_A + N_B$, the linear terms cancel out ($-N + N_A + N_B = 0$):

$$\begin{aligned}\log \frac{N!}{N_A!\,N_B!} &\approx N_A \log N + N_B \log N - N_A \log N_A - N_B \log N_B \\ &= -N_A \log\left(\frac{N_A}{N}\right) - N_B \log\left(\frac{N_B}{N}\right) \\ &= -N_A \log x_A - N_B \log x_B\end{aligned}$$

Therefore,

$$\Delta F = T\Big[N_A \log x_A + N_B \log x_B\Big]$$

Because $x_A < 1$ and $x_B < 1$, their logarithms are negative, implying $\Delta F < 0$.

4. The entropy of mixing is given by the thermodynamic derivative of the free energy:

$$\Delta S_{\text{mix}} = -\left(\frac{\partial \Delta F}{\partial T}\right)_{V,N_A,N_B}$$

Since $\Delta F = T[N_A \log x_A + N_B \log x_B]$, taking the derivative with respect to T simply removes the T:

$$\Delta S_{\text{mix}} = -\Big[N_A \log x_A + N_B \log x_B\Big]$$

This directly satisfies the relation $\Delta F = -T\Delta S_{\text{mix}}$, implying $\Delta U_{\text{mix}} = 0$.

5. **Applicability to dense liquid solutions:** The derivation relies solely on the cancellation of the configurational integral $\int e^{-\beta U} dq^N$. Because we assumed an ideal solution ($u_{AA} = u_{BB} = u_{AB}$), the total potential energy of any spatial configuration is completely oblivious to the A/B label assignments. Therefore, the configurational integral factors out entirely, regardless of how strong or complex the interactions

are. This holds true for strongly interacting dense liquids just as it does for ideal gases.

Proportionality and Independence of T: Because the interactions are symmetric, the mixing process involves exactly zero change in internal energy ($\Delta U_{\text{mix}} = 0$). Mixing is driven entirely by the increase in the number of accessible microstates due to permutations of distinguishable labels (pure combinatorics). Since this combinatorial factor $N!/(N_A!N_B!)$ counts geometric label assignments, it has no thermal dependence, making the entropy of mixing ΔS_{mix} perfectly independent of T. Consequently, via $\Delta F = \Delta U - T\Delta S_{\text{mix}} = -T\Delta S_{\text{mix}}$, the free energy of mixing must be strictly proportional to T.